Lorentz Transformation

Physics 153 – Spring 2027

1. Coordinate Transformations

In special relativity, we postulate that the laws of physics take the same form in all inertial frames, in every direction of space (isotropy), and at all locations and times (homogeneity). Consider two inertial frames, S and S' in uniform relative motion. Assuming the standard orientation, the origins of both frames coincide at $t = t' = 0$ and the frame S' moves with velocity $\vec{v} = v\,\hat{\imath}$ relative to S. Equivalently, the frame S moves with velocity $-\vec{v} = -v\,\hat{\imath}$ with respect to S'. This is because the x axis in each frame points in the same direction in physical space, and if S sees S' moving "forward," then S' must see S moving "backward."

We'd like to derive a rule for transforming the coordinates from one frame to the other. In the process, we'll discover that the *Lorentz transformation* naturally emerges along with a universal constant c that must be empirically determined. It happens to be finite and coincides with the speed of light, but that's not necessary for this discussion. We will not assume that there exists a maximum speed limit, or that the speed of light is constant in all inertial frames. Nonetheless, we can derive the Lorentz transformations from the principle of relativity and by carefully defining what we mean by coordinate transformations from one inertial frame to another.

A coordinate transformation takes the coordinates of frame S given by (x, y, z, t) and maps them to the appropriate coordinates of S' given by (x', y', z', t'). There should exist a "do nothing" transformation that maps the coordinates from one frame to themselves. We call this the *identity transformation*. Coordinate transformations can be done successively, say from $S \to S'$ and then again from $S' \to S''$. The single transformation from $S \to S''$ should be the *composition* of transformations from $S \to S'$ and $S' \to S''$. Each transformation should have an *inverse* that "undoes" the transformation to arrive at the original coordinates. A transformation followed by its inverse is the same as doing nothing, which is the identity transformation.

Lastly, since coordinate transformations are mappings between coordinate systems, their composition is associative. Thus, the set of inertial coordinate transformations forms what mathematicians call a *group* under composition. Henri Poincaré showed that Lorentz transformations form a group [1], and Vladimir Ignatowski essentially used group theory to derive the Lorentz transformation, up to an unspecified universal constant [2].

2. Linear Transformations

Since S and S' are both inertial frames of reference, the coordinate transformation must be a linear map. A linear transformation is the only transformation that will preserve constant-velocity motion. Remember, in an inertial frame of reference, a free particle must move with constant velocity. This constitutes a law of physics, and it must be preserved in all inertial frames. Since the relative velocity is along the x (and x') axis, nothing happens to the perpendicular directions

so we can omit them for brevity (there's a better argument invoking isotropy in case you're interested). Hence, the most general coordinate transformation from S to S' is

$$x' = \alpha(v)\, x + \beta(v)\, t$$
$$t' = \delta(v)\, x + \gamma(v)\, t$$

where $\alpha(v), \beta(v), \gamma(v)$, and $\delta(v)$ are all functions that depend only on the relative velocity. These functions do not depend on x or t due to homogeneity.

It will be slightly easier to work with the transformation in the form of a *matrix equation*:

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \delta & \gamma \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

3. Motion of the Origins

Consider the motion of the origin of S' relative to S. In the S' coordinates, the origin of S' has coordinates $(0, t')$, while in S, the origin of S' has coordinates $(x = vt, t)$. Hence,

$$\begin{pmatrix} 0 \\ t' \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \delta & \gamma \end{pmatrix} \begin{pmatrix} vt \\ t \end{pmatrix}$$

Assuming $t > 0$, it follows that $\beta = -\alpha v$. Analogously, considering the motion of the origin of S relative to S', the transformation maps $(0, t) \mapsto (x' = -vt', t')$. Thus,

$$\begin{pmatrix} -vt' \\ t' \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \delta & \gamma \end{pmatrix} \begin{pmatrix} 0 \\ t \end{pmatrix}$$

From the second equation, $t' = \gamma t$, and from the first, $-vt' = -v\gamma t = \beta t$, or $\beta = -v\gamma$. Combing our results so far, $\alpha = \gamma$ and the transformation matrix has already simplified to

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma & -v\gamma \\ \delta & \gamma \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

4. Reflection of Spatial Coordinates

Consider a reflection of the spatial coordinates such that $x \to -x$ and $x' \to -x'$. Since space is isotropic, the transformation rule should take the same form but with $v \to -v$. Hence,

$$\begin{pmatrix} -x' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma(-v) & v\gamma(-v) \\ \delta(-v) & \gamma(-v) \end{pmatrix} \begin{pmatrix} -x \\ t \end{pmatrix}$$

or,

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} \gamma(-v) & -v\gamma(-v) \\ -\delta(-v) & \gamma(-v) \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

Comparing this to the original transformation rule, we learn that $\gamma(-v) = \gamma(v)$, so $\gamma(v)$ is an even function, and $\delta(-v) = -\delta(v)$, so $\delta(v)$ is an odd function.

5. Inverse Transformation and Identity

There are two ways to go from S' to S. The first is to apply the inverse matrix to S' coordinates:

$$\begin{pmatrix} x \\ t \end{pmatrix} = \frac{1}{\gamma^2 + v\gamma\delta} \begin{pmatrix} \gamma(v) & v\gamma(v) \\ -\delta(v) & \gamma(v) \end{pmatrix} \begin{pmatrix} x' \\ t' \end{pmatrix}$$

The second is to use the transformation rule from S' to S directly with the understanding that S moves with velocity $-v\,\hat{\imath}$ relative to S'. This results in the matrix equation

$$\begin{pmatrix} x \\ t \end{pmatrix} = \begin{pmatrix} \gamma(-v) & v\gamma(-v) \\ \delta(-v) & \gamma(-v) \end{pmatrix} \begin{pmatrix} x' \\ t' \end{pmatrix}$$

The two transformations must match each other (each transformation is unique), so evidently,

$$\gamma^2 + v\gamma\delta = 1$$

This constitutes a constraint relating γ and δ, so in principle, we could write γ in terms of δ and our transformation matrix now depends on just one unknown function.

The transformation must reduce to the identity in the case $v = 0$. Hence, $\gamma(0) = 1$ & $\delta(0) = 0$.

6. Closure

We now invoke the requirement that successive transformations should constitute a single transformation. More precisely, consider first transforming from S to S' and then from S' to a third frame S'' moving with velocity v' relative to S'. The composition of transformations is

$$\begin{pmatrix} x'' \\ t'' \end{pmatrix} = \begin{pmatrix} \gamma(v') & -v'\gamma(v') \\ \delta(v') & \gamma(v') \end{pmatrix} \begin{pmatrix} \gamma(v) & -v\gamma(v) \\ \delta(v) & \gamma(v) \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

The resulting transformation matrix must have the same entries on the diagonal, so

$$\gamma(v')\gamma(v) - v'\gamma(v')\delta(v) = -v\gamma(v)\delta(v') + \gamma(v')\gamma(v)$$

$$\frac{\delta(v)}{v\gamma(v)} = \frac{\delta(v')}{v'\gamma(v')}$$

For nonzero v, this combination of functions must be a *universal constant*, one and the same for all inertial frames. Define this constant as

$$\frac{\delta(v)}{v\gamma(v)} = \kappa$$

where $[\kappa] = 1/[\text{speed}]^2$. From $\gamma^2 + v\gamma\delta = 1$, it follows that

$$\gamma = \frac{1}{\sqrt{1+\kappa v^2}}$$

where we choose the positive root since $\gamma(0) = 1$, not -1. From the closure requirement, we can also derive the velocity addition rule, but I won't do that here.

7. Causal Ordering Principle

Strictly speaking, this section isn't necessary, but I found the argument to be convincing and wanted to share it with you. Feel free to skip forward to section 8 if you don't find this section to be interesting or compelling.

In this section, we show that $\kappa \leq 0$ is the only possibility that preserves *causal ordering*. If event B lies in the "causal future" of event A in one inertial frame, then B cannot precede A in any inertial frame. In other words, if event A is the "cause" and event B is the "effect," then A must come before B in all inertial frames. To the extent that the laws of physics can establish cause-and-effect relationships, we might view this as a natural extension of the relativity principle. Nonetheless, we can show that $\kappa > 0$ allows violations of causal ordering.

Suppose $\kappa > 0$, and let event A, *the cause*, lie at the spacetime origin of both S and S'. Let event B, *the effect*, have coordinates (x_B, t_B) in S, with $t_B > 0$. Under the generalized transformation,

$$x'_B = \frac{x_B - v t_B}{\sqrt{1+\kappa v^2}}, \qquad t'_B = \frac{t_B + \kappa v\, x_B}{\sqrt{1+\kappa v^2}}.$$

To preserve causal ordering, it must be impossible to find any frame S' in which $t'_B < 0$. However, when $x_B < 0$, then whenever $\kappa v|x_B| > t_B$, $t'_B < 0$. Since v can take any positive value and $x_B < 0$ and $t_B > 0$ are otherwise arbitrary, there is nothing preventing the existence of a frame in which *the effect precedes the cause*. Therefore, the causal ordering principle is violated for any $\kappa > 0$, and we must conclude that $\kappa \leq 0$.

At this stage, it is important to clarify what role *causality* (causal ordering principle) is playing in our argument. We have not determined the value of κ from empirical considerations. Instead, we have imposed a constraint on allowed transformations: if event A can influence event B in one inertial frame, then no inertial observer should describe B as occurring before A. This condition restricts the allowable transformations between frames in much the same way that the relativity principle restricts the form of laws themselves.

Causality, in this sense, is not introduced as an ordinary empirical law like Newton's second law or Maxwell's equations. Rather, it serves as a requirement that our theoretical framework must not allow situations in which observers can disagree about the temporal order of cause and effect in cases where influence is physically possible. If such reversals were allowed, the theory would permit scenarios in which effects precede their causes, undermining the ability of the laws to give observer-independent predictions about physical processes.

At this point, logical structure and consistency conditions narrow the options for κ, but they do not decide between them. Only observation can do that. Empirically, we find that κ is negative, and causality is preserved under the transformation rule.

8. Lorentz Transformation

We introduce a new constant c such that $\kappa = -1/c^2$. We call c the *invariant speed* because it has units of speed and takes the same value in all inertial frames of reference. Only experiment or some empirically adequate law of physics can ultimately give the precise value of c and answer whether it is finite or infinite. In principle, any experiment between inertial frames would be sufficient. Multiply the time coordinates by c so the transformation equation takes the following symmetric form:

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} = \frac{1}{\sqrt{1 - v^2/c^2}} \begin{pmatrix} 1 & -v/c \\ -v/c & 1 \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix}$$

Finally, let $\beta = v/c$ and $\gamma = 1/\sqrt{1 - \beta^2}$ so the transformation can be written as

$$\boxed{\begin{pmatrix} x' \\ ct' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix}}$$

Resources

[1] Poincaré, H. (1905). Sur la dynamique de l'électron. *Comptes Rendus de l'Académie des Sciences*, t. 140, p. 1504–1508.

[2] Ignatowsky, W. v. (1910b). Einige allgemeine Bemerkungen zum Relativitätsprinzip. *Physikalische Zeitschrift*. **11**: 972–976.